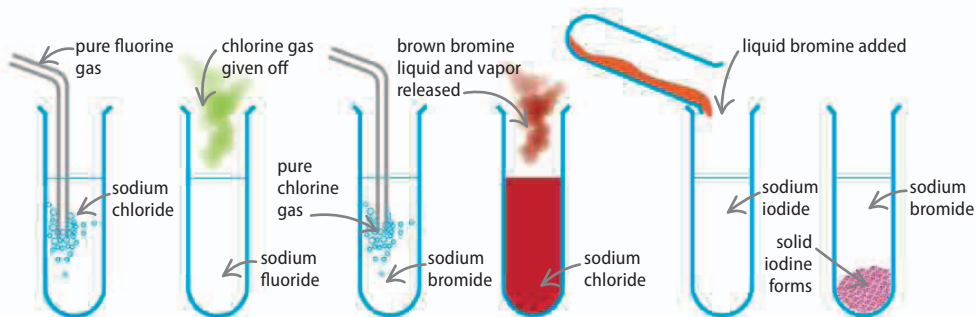


Displacement

Halogens all react in the same way and form similar families of compounds. Therefore, a more highly reactive halogen will displace a less reactive one from its compounds. This can involve the two halogens swapping places in two compounds. When a pure halogen is used, the displaced element is also released in its pure form.



△ Fluorine displaces chlorine

Fluorine displaces chlorine from sodium chloride (NaCl). It will also displace bromine and iodine.

△ Chlorine displaces bromine

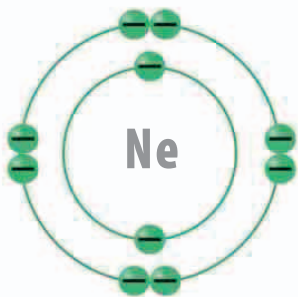
Chlorine displaces bromine from sodium bromide (NaBr). It will also displace iodine but not fluorine.

△ Bromine displaces iodine

Bromine displaces iodine from sodium iodide (NaI). It will not displace chlorine or fluorine.

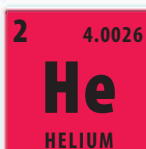
Inert gases

The noble gases form Group 18 of the periodic table. Apart from helium, they all have atoms with eight electrons in their outer shells—a full set. This makes them chemically inactive or inert. In other words, they are noble and do not mix with the other elements, and hardly ever take part in chemical reactions. Their atoms do not form molecules, even with themselves, and all Group 18 elements exist as gases made up of single atoms.



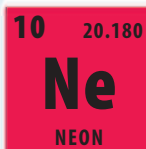
△ Neon atom's shell

Like all noble gases, neon does not bond ionically—it has no spaces to fill in its outer shell. Neon atoms do not share electrons in covalent bonds for the same reason.



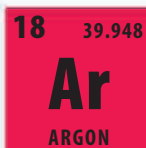
◁ Helium

Helium has just two outer electrons, filling a single shell around the nucleus.



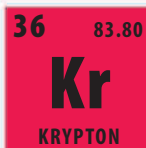
◁ Neon

Discovered in 1898, this gas's name means "the new one."



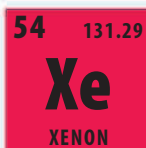
◁ Argon

The most common noble gas on Earth, it forms one percent of the atmosphere.



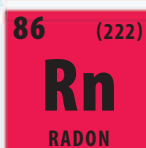
◁ Krypton

Much rarer than neon, this gas's name means "the hidden one."



◁ Xenon

Xenon—"the strange one"—is a dense gas; a balloon of it falls straight down.

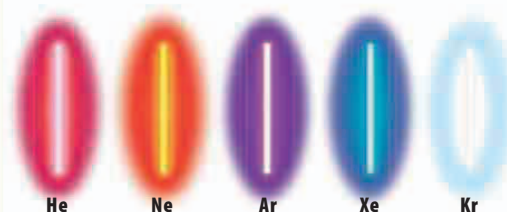


◁ Radon

All radon atoms are radioactive. The gas is formed naturally when uranium in rocks breaks down.

Neon lights

When heated, noble gases glow a specific color. Helium was discovered by its characteristic colors coming from the Sun—and it was named after the Greek word *helios* for Sun. Electrifying noble gases has the same effect, and these are used in gas-discharge lamps—or neon lighting.



△ Glowing gases

In a neon light, electrical current runs through a tube of noble gas. As electrons are ripped off the atoms, they release a certain color of light.

REAL WORLD

Helium balloons

Helium is the second lightest gas in the Universe after hydrogen. Helium balloons float upward in the denser air gases. While hydrogen balloons and airships explode easily, helium balloons cannot burn, so they are safe to use in any size.

